**Depropanizer Column Ver 1 · OTS**

# DC3-V1-ME-DS-001

# Equipment datasheets

**Rev 0 · Issued for teaching**

| | |
|---|---|
| Unit | Depropanizer Column Ver 1 · OTS |
| Document number | DC3-V1-ME-DS-001 |
| Title | Equipment datasheets |
| Discipline | Mechanical |
| Originator / DRAWN | Apinan Soottitantawat |
| Date | 25-OCT-26 |
| Source | simulator model (design snapshot and model runs), epc/epc.json |

## Revision record

| REV | DATE | DESCRIPTION | BY | CHK | APP |
|---|---|---|---|---|---|
| 0 | 25-OCT-26 | Issued for teaching | A. Soottitantawat | | |

***Note.*** *Numbers come from the simulator model (design case) and the assumptions listed; not for construction. Codes are named for the method only; no client, contractor, licensor or vendor is implied.*

## Contents

**Purpose** 3

**C-201 · Depropanizer column** 4

**V-201 · Feed surge drum** 5

**V-202 · Reflux drum** 6

**E-201 · Feed preheater** 7

**E-202 · Overhead condenser** 8

**E-203 · Reboiler** 9

**P-201A/B · Feed pumps** 10

**P-202A/B · Reflux / product pumps** 11

**C-201 · General arrangement (sketch) and nozzle schedule** 12

**Engineering queries raised** 13

# Purpose

One datasheet per item of the C-201 unit, as issued with the enquiry to the vendors. Process data from the HMB DC3-V1-PE-HMB-001, mechanical data from the equipment list DC3-V1-ME-EQL-001; the last sheet is the general-arrangement sketch of C-201 with its nozzle schedule. The right-hand column of each sheet names the source of the value; “assumed” values are design-basis assumptions to be confirmed.

### Design pressure rules

- (1) Vessels and exchangers: design pressure = max(1.10 × maximum operating pressure, maximum operating pressure + 0.17 MPa), rounded up to 0.05 MPa; maximum operating pressure = the highest value over the three HMB cases.
- (2) Vessels holding LPG: not below the bubble (vapour) pressure of the contents at the maximum metal temperature 50 °C (API 2510 practice, assumed).
- (3) Pump discharge systems: not below the shut-off pressure = maximum operating suction pressure + shut-off differential of the pump curve.
- (4) Lower sections of the column: design pressure at the top + column ΔP (PDI-212 high alarm) + static head of liquid to the reboiler return nozzle.
- (5) Utility sides: LP steam 0.70 MPa g, cooling water 0.70 MPa g (assumed system design pressures).
- (6) The PSV set pressure must not exceed the design pressure (MAWP) of the protected equipment (ASME VIII UG-134).

### Design temperature rules

- (a) Design temperature = maximum operating temperature + 28 °C, rounded up to 5 °C.
- (b) Both sides of a steam-heated exchanger: saturation temperature at the steam design pressure (0.70 MPa g → 170 °C), rounded up to 175 °C.
- (c) Uninsulated vessels holding LPG: not below 65 °C (solar gain, assumed).
- (d) Minimum design metal temperature (MDMT) −46 °C for all LPG-containing equipment and piping (auto-refrigeration of propane on depressuring: normal boiling point −42 °C); 0 °C for utility sides (assumed).

### Materials and thickness basis

- SA-516 Gr. 70 plate, allowable stress 138 MPa up to 343 °C (ASME II-D), joint efficiency 0.85 (spot radiography), corrosion allowance 3 mm, mill under-tolerance 0.3 mm (all assumed); impact tested to −46 °C (LPG service).
- Shell thickness ASME VIII Div. 1 UG-27(c)(1) (hoop stress governs), 2:1 ellipsoidal heads UG-32(d), minimum UG-16(b); nominal = next standard plate after the mill under-tolerance; MAWP back-calculated in the corroded condition; hydrotest UG-99(b) 1.3 × design pressure (stress ratio 1).
- Vessel sizes from the model volumes with L/D ≈ 3 and 2:1 heads; exchanger areas = model UA ÷ the Design tab's service U (500 / 700 / 1000 W/m² K for E-201 / E-202 / E-203, assumed).
- External pressure (vacuum), wind, seismic, nozzle loads and local stresses are not checked (query EM-06).

# C-201 · Depropanizer column

Datasheet 1 of 9 · DC3-V1-ME-DS-001 · Rev 0 · Issued for teaching

| | | |
|---|---|---|
| **General** | | |
| Item | C-201 · Depropanizer column | |
| Type | Trayed column, 27 sieve trays (tray type assumed), 2:1 ellipsoidal heads, skirt-supported | |
| Quantity | 1 | |
| Size | 1.7 m ID × 23.0 m T/T; tray spacing 0.61 m; feed tray 18 | model / Design tab |
| Code | ASME BPVC VIII Div. 1 (UG-27, UG-32, UG-99) | |
| **Design conditions** | | |
| Shell — top: operating P / T | 1.700 MPa g / 45.0 °C (max of the cases 1.702 / 45.1) | HMB |
| Shell — top: design P / T / MDMT | 1.90 MPa g / 140 °C / −46 °C | design basis rules |
| Shell — bottom: operating P / T | 1.718 MPa g / 109.0 °C (max of the cases 1.722 / 109.1) | HMB |
| Shell — bottom: design P / T / MDMT | 1.95 MPa g / 140 °C / −46 °C | design basis rules |
| Corrosion allowance | 3 mm | assumed |
| Joint efficiency | 0.85 | assumed (spot RT) |
| Hydrotest (UG-99) | 2.54 MPa g (1.3 × design pressure) | |
| **Materials and construction** | | |
| Shell and heads | SA-516 Gr. 70 (impact tested); trays CS (assumed) | assumed |
| Allowable stress | 138 MPa | assumed (ASME II-D) |
| Shell / head nominal thickness | 18 / 18 mm (required 17.31 mm incl. CA) | Mechanical tab calculator |
| MAWP (corroded, nominal less tolerance) | 2.001 MPa g | |
| Empty weight (estimate) | 23,200 kg — shell + heads + trays 40 kg/m² + 10 % nozzles/internals (estimate) | estimate |
| Insulation | H 50 mm mineral wool (heat conservation), fire-safe on the skirt (assumed) | assumed |
| **Internals** | | |
| Trays | 27 sieve trays (type assumed), spacing 0.61 m, numbered from the bottom; feed on tray 18 | model NT, NF |
| Hydraulics at design / max feed | flood 77 / 89 % rectifying, 74 / 85 % stripping; ΔP 17.9 kPa (PDI-212) | model |
| Sump | 8 m³ to 100 % of LIC-202 range; residence 8.6 min at 50 % (Design tab) | model K.Vsump |
| Skirt | 3.0 m high, fire-proofed (assumed) | assumed |
| **Nozzles (main)** | | |
| N1 | Feed inlet (tray 18, with distributor) — 6" (6"-P-203) at 14.67 m above BTL | |
| N2 | Reflux inlet (above tray 27) — 4" (4"-P-209) at 20.16 m above BTL | |
| N3 | Overhead vapour outlet — 8" (8"-P-204) at 23.39 m above BTL | |
| N4 | Bottoms outlet to LV-202 — 4" (4"-P-210) at -0.42 m above BTL | |
| N5 | Reboiler draw-off to E-203 — 8" (8"-P-211) at 0.3 m above BTL | |
| N6 | Reboiler return from E-203 (below tray 1) — 10" (10"-P-212) at 3.45 m above BTL | |
| M1 | Manway 24" (above tray 27) — 24" at 20.76 m above BTL | |
| M2 | Manway 24" (feed zone, tray 19) — 24" at 15.28 m above BTL | |
| M3 | Manway 24" (tray 9) — 24" at 9.18 m above BTL | |
| M4 | Manway 24" (sump) — 24" at 2.4 m above BTL | |
| **Process data** | | |
| Overhead vapour | 40,798 kg/h at 45.0 °C (max feed 47,608) | HMB stream 4 |
| Bottoms | 15,619 kg/h at 109.0 °C | HMB stream 8 |
| Relief | PSV-201 set 1.80 MPa g on the overhead line; Design-tab orifice API 'L' (1841 mm²) = L on the P&ID rev D; DC3-V1-PE-RLB-001 needs M (2323 mm²), see ER-01. Relieving temperature 77.7 °C (no reflux) / 124 °C (fire on C-201) vs 75 °C design of the overhead system, see ER-04 | Relief tab |
| **Notes** | | |
| | Diameter and height from the Design tab sizing; design pressure rule (1) at the top, rule (4) at the bottom; PSV-201 on the overhead line set 1.80 MPa g. | |

# V-201 · Feed surge drum

Datasheet 2 of 9 · DC3-V1-ME-DS-001 · Rev 0 · Issued for teaching

| | | |
|---|---|---|
| **General** | | |
| Item | V-201 · Feed surge drum | |
| Type | Horizontal drum, 2:1 ellipsoidal heads | |
| Quantity | 1 | |
| Size | 2.1 m ID × 6.5 m T/T (24.8 m³ from the model, L/D ≈ 3 assumed) | model / Design tab |
| Code | ASME BPVC VIII Div. 1 (UG-27, UG-32, UG-99) | |
| **Design conditions** | | |
| Shell: operating P / T | 0.400 MPa g / −1.0 °C (max of the cases 0.400 / −1.0) | HMB |
| Shell: design P / T / MDMT | 1.35 MPa g / 65 °C / −46 °C | design basis rules |
| Corrosion allowance | 3 mm | assumed |
| Joint efficiency | 0.85 | assumed (spot RT) |
| Hydrotest (UG-99) | 1.76 MPa g (1.3 × design pressure) | |
| **Materials and construction** | | |
| Shell and heads | SA-516 Gr. 70 (impact tested) | assumed |
| Allowable stress | 138 MPa | assumed (ASME II-D) |
| Shell / head nominal thickness | 16 / 16 mm (required 15.20 mm incl. CA) | Mechanical tab calculator |
| MAWP (corroded, nominal less tolerance) | 1.405 MPa g | |
| Empty weight (estimate) | 7,400 kg — shell + heads + 10 % (estimate) | estimate |
| Insulation | AC 40 mm anti-condensation (operates at −1 °C, assumed) | assumed |
| **Process data** | | |
| Liquid | feed LPG, 31,809 kg/h at −1.0 °C, bubble pressure 1.30 MPa g at 50 °C | HMB stream 1 |
| Volume / hold-up | 24.8 m³; 12.4 min at 50 % level and design feed | model / Design tab |
| Level instruments | LG-201 gauge glass (rev A); LT-201 / LIC-201 in rev B / rev C | P&ID |
| **Notes** | | |
| | Design pressure set by rule (2): feed bubble pressure at 50 °C 1.30 MPa g. No PSV shown on the P&ID (note 11, HOLD). | |

# V-202 · Reflux drum

Datasheet 3 of 9 · DC3-V1-ME-DS-001 · Rev 0 · Issued for teaching

| | | |
|---|---|---|
| **General** | | |
| Item | V-202 · Reflux drum | |
| Type | Horizontal drum, 2:1 ellipsoidal heads | |
| Quantity | 1 | |
| Size | 1.8 m ID × 5.6 m T/T (15.7 m³ from the model) | model / Design tab |
| Code | ASME BPVC VIII Div. 1 (UG-27, UG-32, UG-99) | |
| **Design conditions** | | |
| Shell: operating P / T | 1.680 MPa g / 39.4 °C (max of the cases 1.691 / 41.8) | HMB |
| Shell: design P / T / MDMT | 1.90 MPa g / 75 °C / −46 °C | design basis rules |
| Corrosion allowance | 3 mm | assumed |
| Joint efficiency | 0.85 | assumed (spot RT) |
| Hydrotest (UG-99) | 2.47 MPa g (1.3 × design pressure) | |
| **Materials and construction** | | |
| Shell and heads | SA-516 Gr. 70 (impact tested) | assumed |
| Allowable stress | 138 MPa | assumed (ASME II-D) |
| Shell / head nominal thickness | 20 / 18 mm (required 17.77 mm incl. CA) | Mechanical tab calculator |
| MAWP (corroded, nominal less tolerance) | 1.906 MPa g | |
| Empty weight (estimate) | 6,700 kg — shell + heads + 10 % (estimate) | estimate |
| Insulation | None (personnel protection not required below 60 °C) | assumed |
| **Process data** | | |
| Liquid | propane, 40,710 kg/h (reflux + product) at 39.4 °C | HMB streams 5 + 6 |
| Vapour outlet | vent to fuel gas 72.5 Nm³/h (max feed 437) | HMB stream 7 |
| Volume / hold-up | 15.7 m³; 5.8 min at 50 % level | model / Design tab |
| Level instruments | LT-203 / LIC-203 and LS-203 / LSHH-203 on separate nozzles | P&ID note 6 |
| **Notes** | | |
| | Protected by PSV-201 through E-202 (no isolation valve between C-201 and V-202). | |

# E-201 · Feed preheater

Datasheet 4 of 9 · DC3-V1-ME-DS-001 · Rev 0 · Issued for teaching

| | | |
|---|---|---|
| **General** | | |
| Item | E-201 · Feed preheater | |
| Type | Shell & tube, TEMA BEU (feed in the U-tubes, LP steam in the shell) — assumed | assumed |
| Code | ASME VIII Div. 1 + TEMA class R | |
| Size | 90 m² (= model UA 45 kW/K ÷ U 500 W/m² K assumed); duty 3.50 MW | model UA / assumed U |
| **Thermal** | | |
| Duty | 3,503 kW design / 4,074 kW max feed | HMB |
| UA (model) | 45 kW/K (on T_sat − mean feed T) | K.UAp |
| U (service, assumed) / area | 500 W/m² K / 90 m² | Design tab U |
| Design-tab area | 82.5 m² (LMTD 84.9 K) | Design tab |
| **Shell side** | | |
| Fluid | LP steam (condensing) | |
| Flow | 5,817 kg/h (max feed 6,902) | HMB stream 9 |
| Chest pressure | 148 kPa g (max 260, turndown 6.7) | model PI (P.Psp) |
| **Tube side** | | |
| Fluid | Feed LPG, partly vaporized | |
| Flow | 31,809 kg/h | HMB stream 2 |
| Temperature in / out | −1.0 / 75.0 °C, outlet 67 mol % vapour | HMB streams 2, 3 |
| **Design conditions** | | |
| Shell (LP steam): operating P / T | 0.148 MPa g / 127.5 °C (max of the cases 0.260 / 140.2) | HMB |
| Shell (LP steam): design P / T / MDMT | 0.70 MPa g / 175 °C / 0 °C | design basis rules |
| Tubes (feed): operating P / T | 2.363 MPa g / 75.0 °C (max of the cases 2.563 / 75.0) | HMB |
| Tubes (feed): design P / T / MDMT | 2.85 MPa g / 175 °C / −46 °C | design basis rules |
| **Materials** | | |
| Materials | Shell SA-516 Gr. 70; tubes SA-179 seamless CS (assumed) | assumed |
| Insulation | H 50 mm (assumed) | assumed |
| **Notes** | | |
| | Tube side on the P-201 discharge: design pressure rule (3) (shut-off 2.72 MPa g). | |

# E-202 · Overhead condenser

Datasheet 5 of 9 · DC3-V1-ME-DS-001 · Rev 0 · Issued for teaching

| | | |
|---|---|---|
| **General** | | |
| Item | E-202 · Overhead condenser | |
| Type | Shell & tube, TEMA AES (condensing propane in the shell, cooling water in the tubes) — assumed | assumed |
| Code | ASME VIII Div. 1 + TEMA class R | |
| Size | 290 m² (= model clean UA 203 kW/K ÷ U 700 W/m² K assumed); duty 3.96 MW | model UA / assumed U |
| **Thermal** | | |
| Duty | 3,956 kW design / 4,547 kW max feed | HMB |
| UA (model, clean) | 203 kW/K; working 168.7 kW/K with 16.9 % blanketed by non-condensables | K.UAc, P.a |
| U (service, assumed) / area | 700 W/m² K / 290 m² | Design tab U |
| Design-tab area | 291.3 m² installed (242.1 m² working, LMTD 23.43 K) | Design tab |
| **Shell side** | | |
| Fluid | Overhead propane vapour (condensing) + non-condensables | |
| Flow | 40,798 kg/h (max feed 47,608) | HMB stream 4 |
| Condensing temperature | 44.0 °C at 1.680 MPa g; liquid out 39.4 °C | model |
| **Tube side** | | |
| Fluid | Cooling water | |
| Flow | 330 m³/h | HMB stream 11 |
| Temperature in / out | 15.0 / 25.3 °C | HMB stream 11 |
| **Design conditions** | | |
| Shell (overhead): operating P / T | 1.700 MPa g / 45.0 °C (max of the cases 1.702 / 45.1) | HMB |
| Shell (overhead): design P / T / MDMT | 1.90 MPa g / 75 °C / −46 °C | design basis rules |
| Tubes (cooling water): operating P / T | 0.450 MPa g / 25.3 °C (max of the cases 0.450 / 26.8) | HMB |
| Tubes (cooling water): design P / T / MDMT | 0.70 MPa g / 75 °C / 0 °C | design basis rules |
| **Materials** | | |
| Materials | Shell SA-516 Gr. 70; tubes SA-179 CS, CW side CA 3 mm (assumed) | assumed |
| Insulation | None | assumed |
| **Notes** | | |
| | CW side design pressure 0.70 MPa g is below 10/13 of the shell side — tube-rupture case for the Relief tab. | |

# E-203 · Reboiler

Datasheet 6 of 9 · DC3-V1-ME-DS-001 · Rev 0 · Issued for teaching

| | | |
|---|---|---|
| **General** | | |
| Item | E-203 · Reboiler | |
| Type | Vertical thermosyphon, TEMA BEM (process in the tubes, LP steam in the shell) — assumed | assumed |
| Code | ASME VIII Div. 1 + TEMA class R | |
| Size | 73 m² (= model UA 73 kW/K ÷ U 1000 W/m² K assumed); duty 2.24 MW | model UA / assumed U |
| **Thermal** | | |
| Duty | 2,239 kW design / 2,573 kW max feed | HMB |
| UA (model) | 73 kW/K (on T_sat(chest) − T_bottom) | K.UAr |
| U (service, assumed) / area | 1000 W/m² K / 73 m² | Design tab U |
| Design-tab area | 74 m² (ΔT 30.64 K) | Design tab |
| **Shell side** | | |
| Fluid | LP steam (condensing) | |
| Flow | 3,790 kg/h (max feed 4,389) | HMB stream 10 |
| Chest pressure | 254 kPa g (max 304) | model (P.Psr, PI-205) |
| **Tube side** | | |
| Fluid | C-201 bottoms (boiling), thermosyphon | |
| Boil-up | 520 kmol/h vapour; circulation at 30 % vaporized (assumed) | model P.V0 |
| Temperature | 109.0 °C | model |
| **Design conditions** | | |
| Shell (LP steam): operating P / T | 0.254 MPa g / 139.7 °C (max of the cases 0.304 / 144.3) | HMB |
| Shell (LP steam): design P / T / MDMT | 0.70 MPa g / 175 °C / 0 °C | design basis rules |
| Tubes (C-201 bottoms): operating P / T | 1.736 MPa g / 109.0 °C (max of the cases 1.740 / 109.1) | HMB |
| Tubes (C-201 bottoms): design P / T / MDMT | 1.95 MPa g / 175 °C / −46 °C | design basis rules |
| **Materials** | | |
| Materials | Shell SA-516 Gr. 70; tubes SA-179 seamless CS (assumed) | assumed |
| Insulation | H 50 mm (assumed) | assumed |
| **Notes** | | |
| | Circulation rate is not modelled; 30 % mass vaporized at the outlet assumed for the line sizing. | |

# P-201A/B · Feed pumps

Datasheet 7 of 9 · DC3-V1-ME-DS-001 · Rev 0 · Issued for teaching

| | | |
|---|---|---|
| **General** | | |
| Item | P-201A/B · Feed pumps | |
| Type | Centrifugal, API 610 between-bearings multistage (head ≈ 361 m) — type assumed; A motor, B steam turbine | assumed |
| Code | API 610 | |
| Driver | P-201A motor, P-201B steam turbine (diverse drive) | P&ID note 7 |
| **Liquid** | | |
| Fluid | Feed LPG (C3 / C4 pseudo-binary) | |
| Pumping temperature | −1.0 °C | HMB |
| Density | 548.5 kg/m³ | design basis |
| Vapour pressure | 414 kPa a | model VLE |
| Viscosity | 0.10 mPa s | assumed |
| **Performance** | | |
| Flow normal / max feed / rated | 60.0 / 70.0 / 77.0 m³/h | HMB; rated = 110 % of max |
| Suction / discharge pressure (design) | 423 / 2,363 kPa g | model |
| Differential pressure normal / rated | 1,940 / 1,707 kPa (360 / 317 m) | model pump curve |
| Shut-off / run-out | 2,723 kPa g / 152 m³/h | model curve |
| NPSHa (model / liquid at bubble point) | 20.2 / 3.9 m | eproc tab |
| NPSHr (assumed) | 3 m + 1 m margin | assumed |
| Efficiency | 65 % | assumed |
| Absorbed power normal / max / rated | 49.7 / 54.1 / 56.2 kW | flow × ΔP ÷ η |
| Motor rating | 75 kW, 400 V (API 610 margin 10 %) | Electrical tab |
| **Design conditions** | | |
| Casing: operating P / T | 2.363 MPa g / −1.0 °C (max of the cases 2.563 / −1.0) | HMB |
| Casing: design P / T / MDMT | 2.85 MPa g / 65 °C / −46 °C | design basis rules |
| **Construction** | | |
| Materials | CS casing, impact tested; 12 % Cr internals (assumed) | assumed |
| Mechanical seal | API 682 arrangement 2 (dual, unpressurised) — flashing LPG | assumed |
| **Notes** | | |
| | Pump curve and system from the model (H0 2300 kPa, a 0.1 kPa/(m³/h)²). P-201B: steam turbine drive (diverse drive, P&ID note 7). | |

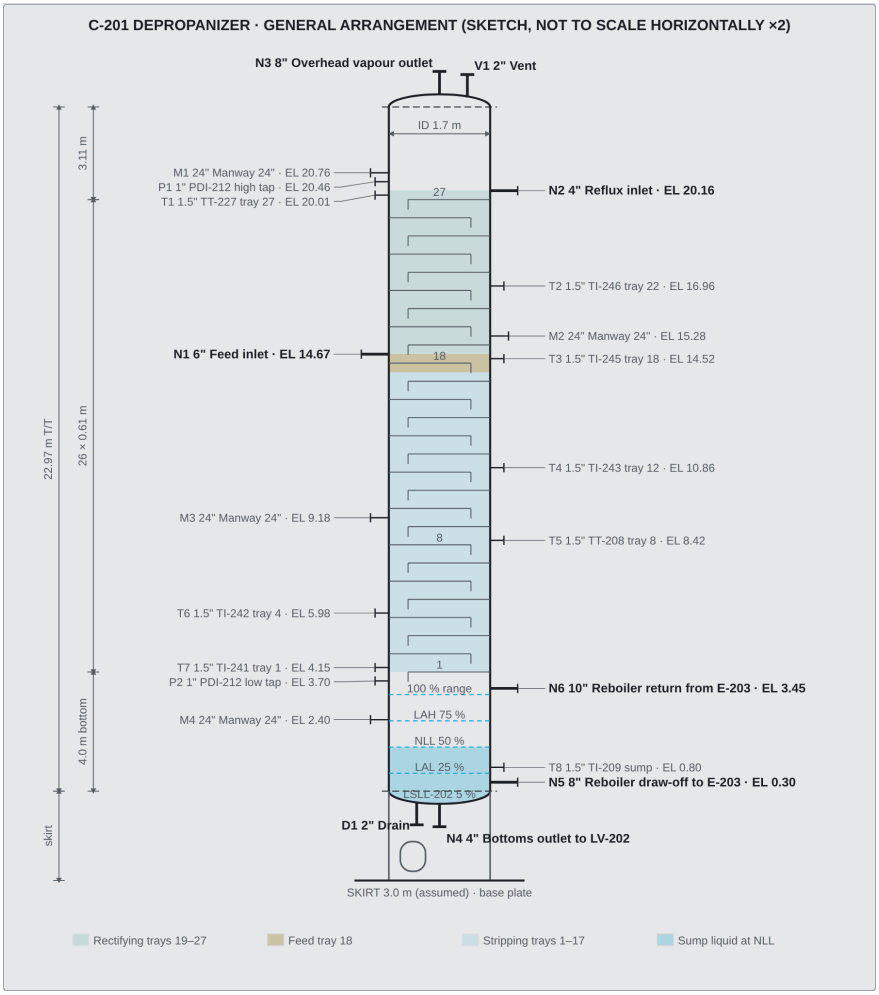
# P-202A/B · Reflux / product pumps

Datasheet 8 of 9 · DC3-V1-ME-DS-001 · Rev 0 · Issued for teaching

| | | |
|---|---|---|
| **General** | | |
| Item | P-202A/B · Reflux / product pumps | |
| Type | Centrifugal, API 610 OH2 single stage — type assumed; motor drives | assumed |
| Code | API 610 | |
| Driver | electric motors (A and B) | P&ID note 7 |
| **Liquid** | | |
| Fluid | Propane (98.5 mol % C3 and lighter) | |
| Pumping temperature | 39.4 °C | HMB |
| Density | 471.7 kg/m³ | design basis |
| Vapour pressure | 1,622 kPa a | model VLE |
| Viscosity | 0.10 mPa s | assumed |
| **Performance** | | |
| Flow normal / max feed / rated | 81.1 / 93.2 / 102.5 m³/h | HMB; rated = 110 % of max |
| Suction / discharge pressure (design) | 1,710 / 2,262 kPa g | model |
| Differential pressure normal / rated | 551 / 462 kPa (119 / 100 m) | model pump curve |
| Shut-off / run-out | 2,421 kPa g / 176 m³/h | model curve |
| NPSHa (model / liquid at bubble point) | 40.5 / 6.1 m | eproc tab |
| NPSHr (assumed) | 3 m + 1 m margin | assumed |
| Efficiency | 60 % | assumed |
| Absorbed power normal / max / rated | 20.7 / 21.7 / 22.0 kW | flow × ΔP ÷ η |
| Motor rating | 30 kW, 400 V (API 610 margin 25 %) | Electrical tab |
| **Design conditions** | | |
| Casing: operating P / T | 2.262 MPa g / 39.4 °C (max of the cases 2.355 / 41.8) | HMB |
| Casing: design P / T / MDMT | 2.60 MPa g / 70 °C / −46 °C | design basis rules |
| **Construction** | | |
| Materials | CS casing, impact tested; 12 % Cr internals (assumed) | assumed |
| Mechanical seal | API 682 arrangement 2 (dual, unpressurised) — flashing LPG | assumed |
| **Notes** | | |
| | Pump curve from the model (H0R 700 kPa, a 0.0226). PDSL-230 auto-starts P-202B. | |

# C-201 · General arrangement (sketch) and nozzle schedule

Datasheet 9 of 9. Elevations above the bottom tangent line (BTL); tray k at 4.0 + (k − 1) × 0.61 m; skirt 3.0 m (assumed); levels from the model sump volume 8 m³ (0–100 % of LIC-202 range). Horizontal scale ×2.




| Mark | Service | NPS | EL m |
|---|---|---|---|
| N1 | Feed inlet (tray 18, with distributor) | 6" | 14.67 |
| N2 | Reflux inlet (above tray 27) | 4" | 20.16 |
| N3 | Overhead vapour outlet | 8" | 23.39 |
| N4 | Bottoms outlet to LV-202 | 4" | −0.42 |
| N5 | Reboiler draw-off to E-203 | 8" | 0.30 |
| N6 | Reboiler return from E-203 (below tray 1) | 10" | 3.45 |
| M1 | Manway 24" (above tray 27) | 24" | 20.76 |
| M2 | Manway 24" (feed zone, tray 19) | 24" | 15.28 |
| M3 | Manway 24" (tray 9) | 24" | 9.18 |
| M4 | Manway 24" (sump) | 24" | 2.40 |
| T1 | TT-227 thermowell, tray 27 | 1.5" | 20.01 |
| T2 | TI-246 thermowell, tray 22 | 1.5" | 16.96 |
| T3 | TI-245 thermowell, tray 18 | 1.5" | 14.52 |
| T4 | TI-243 thermowell, tray 12 | 1.5" | 10.86 |
| T5 | TT-208 thermowell, tray 8 (control tray) | 1.5" | 8.42 |
| T6 | TI-242 thermowell, tray 4 | 1.5" | 5.98 |
| T7 | TI-241 thermowell, tray 1 | 1.5" | 4.15 |
| T8 | TI-209 thermowell, sump | 1.5" | 0.80 |
| L1/L2 | LT-202 level bridle (control) | 2" | −0.20 / 3.30 |
| L3/L4 | LS-202 level (trip, separate nozzles) | 2" | −0.25 / 3.30 |
| P1 | PDI-212 high tap (above tray 27) | 1" | 20.46 |
| P2 | PDI-212 low tap (sump vapour space) | 1" | 3.70 |
| V1 | Vent (top head) | 2" | 23.29 |
| D1 | Drain (bottom head) | 2" | −0.42 |

Sump levels: LSLL-202 (trip) 5 % at −0.11 m, LAL 25 % at 0.60 m, NLL (LIC-202 SP) 50 % at 1.48 m, LAH 75 % at 2.36 m, 100 % of range 100 % at 3.24 m (above BTL; negative = in the bottom head).

# Engineering queries raised

| No. | Subject | Query | Proposed answer | Status |
|---|---|---|---|---|
| EM-01 | Minimum design metal temperature and material | Depressuring LPG auto-refrigerates to about −42 °C; MDMT −46 °C was applied to C-201, V-201, V-202, E-202 shell, the pumps and the hydrocarbon piping. Plain SA-516-70 is not exempt from impact testing at this MDMT (UCS-66). | Specify SA-516-70N impact tested at −46 °C (and LT piping classes A1L / B1L); confirm the depressuring end temperature in the Relief tab. | OPEN |
| EM-02 | V-201 design pressure and overpressure protection | Rule (2) sets V-201 at 1.35 MPa g (feed bubble pressure 1.30 MPa g at 50 °C) while it operates at 0.40 MPa g; the P&ID shows no PSV on V-201 (note 11). | Add a PSV on V-201 set at 1.35 MPa g (fire case and upstream blocked-in); confirm the upstream maximum pressure. | OPEN |
| EM-03 | E-202 tube rupture (cooling-water side) | The CW side of E-202 is designed for 0.70 MPa g, below 10/13 of the shell design pressure (1.46 MPa g): a tube rupture would overpressure the cooling-water system with propane. | Raise the CW side to 1.50 MPa g or provide tube-rupture relief (TSV-216 recommended in the Safety tab) — Relief tab to size. | OPEN |
| EM-04 | E-201 / E-203 steam side vs process side | The LP-steam shells (0.70 MPa g) are below 10/13 of the tube-side design pressure (E-201 2.85, E-203 1.95 MPa g): a tube rupture sends hydrocarbon into the steam / condensate system. | Tube-rupture case for both shells in the Relief tab; hydrocarbon detection on the condensate return (recommended addition). | OPEN |
| EM-05 | Pump discharge design pressure basis | P-201 discharge items (E-201 tubes, 4"-P-202, 6"-P-203) use the shut-off pressure at the normal suction pressure (2.72 MPa g). If V-201 can reach its design pressure 1.35 MPa g, the shut-off rises to 3.67 MPa g. | Decide whether the maximum suction pressure is the V-201 operating or design pressure; with the latter the discharge piping moves to class 300 at 3.70 MPa g. | OPEN |
| EM-06 | External pressure, wind and seismic not checked | C-201 (1.7 m × 23.0 m) is checked for internal pressure only. Steam-out and cooling of a blocked-in column can pull vacuum; wind and seismic loads set the bottom-course and skirt thickness of tall columns. | Design C-201, V-201 and V-202 for full vacuum (UG-28) or provide vacuum protection; civil/structural to run wind and seismic checks of C-201 and its skirt. | OPEN |